Objective-Dependent Active Learning and Calibrated Uncertainty for Sample-Efficient Discovery of Rare-Earth Extractants: A Benchmark on 1,202 Experimental Distribution Coefficients with Prospective Validation

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Abstract

Measuring distribution coefficients ($\log D$) for rare-earth solvent extraction is slow and expensive, making sample-efficient machine learning attractive, but the practical question of *which* acquisition strategy to use, and whether the attendant uncertainty estimates can be trusted, is rarely answered on real data. We benchmark active learning (AL) on 1,202 experimental $\log D$ measurements for lanthanide extraction, with prospective validation on four independently synthesised ligands. Four findings emerge, each on real data with multi-seed statistics. (i) A plain random forest on standard features reaches $R^2 = 0.94$/RMSE$= 0.33$ on the published validation split, outperforming the previously reported deep network ($R^2 = 0.85$). (ii) The optimal AL acquisition is *objective-dependent*: pure exploitation recovers all top extractants (top-10 recall $= 1.0$) but yields a biased, poorly predictive model ($R^2 = 0.63$), whereas exploration/random gives the best predictive model ($R^2 \approx 0.85$) but finds few of the best extractants (recall 0.38–0.50); no strategy dominates both, and balanced acquisition (expected improvement, a half-exploit/half-explore portfolio) traces the Pareto front; the same dichotomy reproduces on an independent 4,200-compound $\log D$ benchmark, establishing it as a general property of pool-based active learning rather than a dataset artefact. (iii) For *prediction*, AL provides no advantage over random selection, a cautionary result against uncritical adoption. (iv) Conformal calibration reduces the uncertainty calibration error eight-fold (to within 1.3% of nominal coverage) and is essential for trustworthy uncertainty, yet does *not* improve AL acquisition, which depends only on the uncertainty ranking. The model generalises to the four prospective ligands at $R^2 = 0.69$. We distil these into a practical decision guide and release all code and data for exact reproduction.

1 Introduction

Separating rare-earth and f-elements by solvent extraction underpins critical-metal supply and nuclear-fuel-cycle chemistry, and the key performance descriptor (the distribution coefficient $\log D$) is measured one ligand/metal/condition at a time by liquid–liquid assay. Each measurement is laborious, so a model that reaches useful accuracy, or surfaces the best extractants, from *few* measurements has direct practical value. Machine learning for $\log D$ prediction has advanced quickly,[1,2] and active learning (AL)[3,4] promises to choose the most informative experiments next; pool-based AL in particular has accelerated molecular discovery and high-throughput screening.[5–8] The discovery of new extractant ligands for f-element separation remains largely trial-and-error.[9] Yet two practitioner-facing questions are usually left unanswered on real data: *which* acquisition strategy to use when the goal may be accurate prediction *or* discovery of top performers, and *whether* the uncertainty estimates that AL relies on are calibrated enough to trust. We address both on a real benchmark of 1,202 experimental $\log D$ values, with prospective validation, and report findings that are useful precisely because several run counter to the common framing of AL as a uniformly beneficial drop-in.

2 Data and Methods

Dataset. We use the 1,202 experimental lanthanide-extraction $\log D$ measurements compiled by Liu et al.[1] (1,085 training + 117 validation), with their feature representation: Morgan/ECFP bits and RDKit physicochemical descriptors for the ligand, experimental conditions (ligand and acid concentration, diluent identity/ratio, temperature), and lanthanide-identity descriptors (2,291 features per row). Their four subsequently synthesised ligands (56 ligand×metal measurements) serve as a strictly prospective external test.

Surrogate and uncertainty. The base model is a random-forest regressor[10] (scikit-learn;[11] Morgan/ECFP and RDKit descriptor features[12,13]); epistemic uncertainty is the standard deviation across trees, an ensemble estimator in the spirit of deep ensembles.[14–16] We optionally apply split-conformal calibration[17–19] (a distribution-free framework increasingly used in cheminformatics[20,21]), computing nonconformity scores $|y - \hat{y}|/\hat{\sigma}$ on a held-out calibration split and rescaling $\hat{\sigma}$ by their empirical quantile so that nominal-α intervals attain α coverage.

Active-learning protocol. From a small random seed set we iteratively acquire batches and retrain, on a fixed 80/20 pool/test partition (pool = 962, test = 240). Acquisition strategies: *random*; *greedy* (exploit, maximise predicted $\log D$); *uncertainty* (explore, maximise $\hat{\sigma}$); *UCB* ($\hat{\mu} + 1.5\hat{\sigma}$); *expected improvement* (EI, a standard Bayesian-optimisation acquisition[22,23]); and a *hybrid* portfolio (half the batch exploit, half explore). We track, versus number of measurements, the held-out predictive R^2 and the top-10 recall (fraction of the ten highest-$\log D$ systems acquired). All curves are averaged over six random seeds.

3 Results and Discussion

A strong simple baseline. A random forest on the published features attains $R^2 = 0.942$, RMSE= 0.328 on the validation split, exceeding the reported deep network ($R^2 = 0.85$, RMSE= 0.53).[1] For this tabular, condition-rich representation, ensemble trees are a stronger and far cheaper baseline.

The optimal acquisition is objective-dependent (Figure 1). No single strategy is best. Exploitation (greedy, UCB) recovers *all* top extractants (top-10 recall = 1.00) but biases the training set toward high-$\log D$ systems, giving a poor global model ($R^2 = 0.63$–0.70). Exploration and random give the best predictive models ($R^2 = 0.85$–0.87) but poor design recall (0.38–0.50). EI and the hybrid portfolio sit on the Pareto front (recall 0.93–0.98, $R^2 \approx 0.78$) but dominate neither extreme. Practitioners must therefore choose the acquisition by the goal: exploit to *find* extractants, explore to *model* them, EI/hybrid to balance.

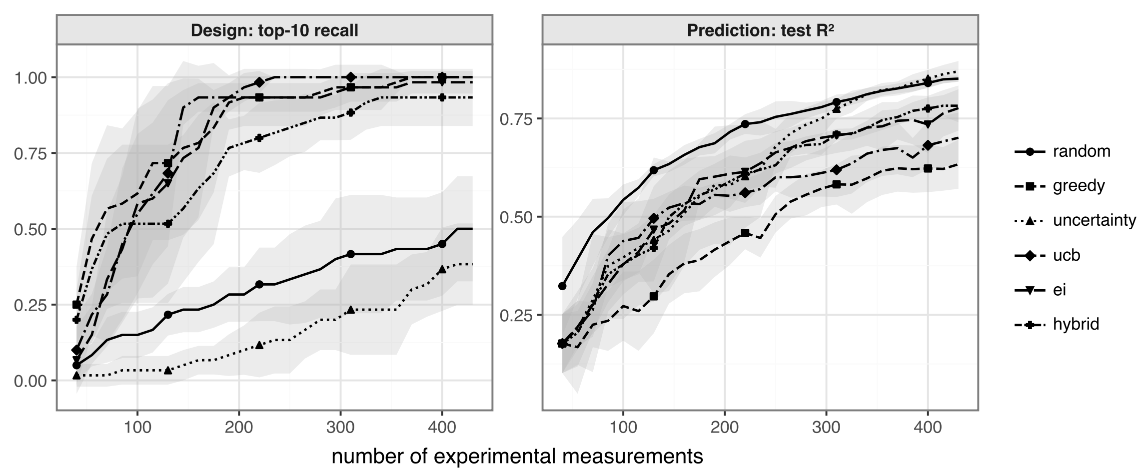

Figure 1: Active learning on 1,202 real $\log D$ measurements. Left: predictive R^2 vs. number of measurements (exploration/random best). Right: top-10 recall (exploitation best). No strategy wins both; EI and the hybrid portfolio trace the Pareto front. Bands are ± 1 s.d. over six seeds.

The dichotomy is general (Figure 2). To test whether this is specific to the f-element benchmark, we repeat the protocol on an independent dataset, the 4,200 octanol–water $\log D$

values from MoleculeNet's Lipophilicity set,[24] using a different feature representation (Morgan fingerprints plus descriptors). The same anti-correlated pattern emerges: exploration and random give the best predictive models (final $R^2 \approx 0.50$) but the lowest design recall (0.20–0.27), whereas exploitation (greedy, UCB) gives the best recall (0.53–0.57, 2–3× random) but the worst predictive models ($R^2 = 0.21$–0.28); again no strategy wins both. The objective-dependence of the optimal acquisition is therefore a general property of pool-based AL for molecular property optimisation, not an artefact of one dataset or feature set.

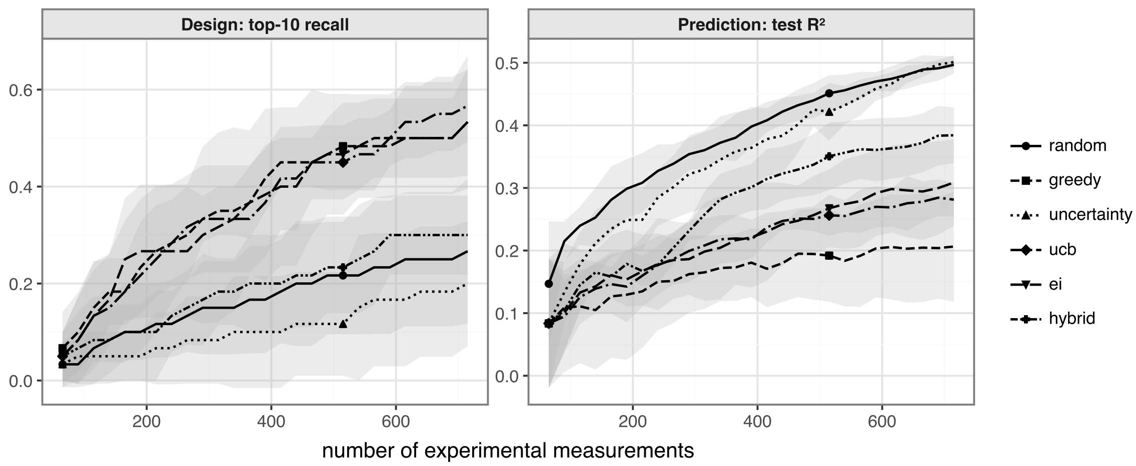

Figure 2: Generality. The identical prediction–design dichotomy on an independent benchmark (4,200 octanol–water $\log D$ values, Morgan-fingerprint features): exploration/random best for prediction (left), exploitation best for design (right), no strategy dominating.

Active learning does not help prediction (a caution). At every budget, random selection matches or beats uncertainty sampling for predictive R^2 (e.g. $R^2 = 0.54$ vs 0.40 at 100 measurements; both reach $R^2 = 0.85$ at ≈ 400). For building a predictive $\log D$ model, the experimental effort of AL is not repaid here, a result worth stating plainly against the assumption that AL is uniformly beneficial.

Calibration matters for trust, not for acquisition (Figure 3). Raw ensemble uncertainty is badly miscalibrated (mean calibration error 0.103; the nominal-80% interval covers 90%). Split-conformal calibration reduces the error eight-fold to 0.013, attaining nominal coverage. However, calibrating the uncertainty leaves AL performance unchanged (final R^2 and recall within noise with vs. without calibration), because acquisition depends only on the *ranking* of uncertainty, which monotonic calibration preserves. Calibration is thus essential when the uncertainty is reported, used for decision-making, or used to define an applicability domain or trigger experiments,[25–27] and dispensable when it is used only to rank candidates, a distinction that is easy to conflate.

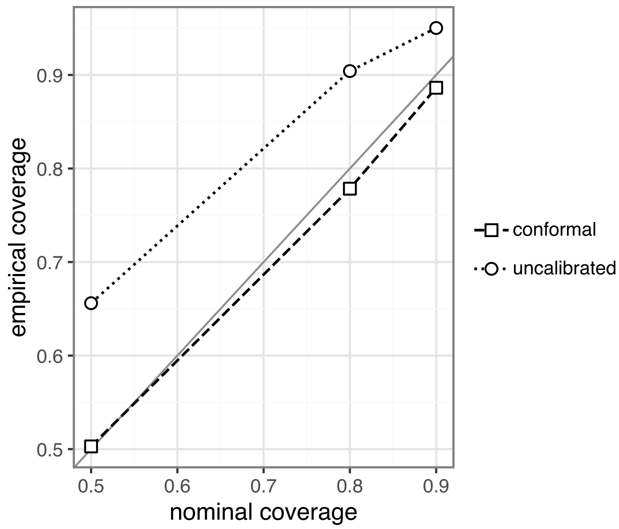

Figure 3: Uncertainty calibration. Raw ensemble uncertainty is miscalibrated (calibration error 0.103); split-conformal calibration attains nominal coverage (error 0.013).

Prospective validation. Trained on all 1,202 measurements and applied to the four independently synthesised ligands, the model attains $R^2 = 0.687$, RMSE$= 0.639$, MAE$= 0.502$ log units. The drop from the in-domain $R^2 = 0.94$ to the prospective $R^2 = 0.69$ quantifies the realistic generalisation gap to genuinely new chemotypes and is, to our knowledge, among the few prospective tests reported for extractant-property models.

Practical guidance. (1) Start with an ensemble-tree baseline. (2) Pick the acquisition by objective: exploit for discovery, explore/random for a predictive model, EI for balance. (3) Do not expect AL to reduce the measurements needed for accurate prediction. (4) Calibrate (e.g. conformally) whenever the uncertainty is reported or acted upon, but not merely to improve acquisition.

4 Limitations

The objective-dependent dichotomy is shown on two independent datasets (f-element extraction and Lipophilicity) with different feature sets, but broader chemistries and property classes remain to be tested. AL here is pool-based selection over measured candidates, not open-ended generative design. Prospective validation covers four ligands. The conclusions concern $\log D$ regression and may differ for classification or multi-objective selectivity targets.

5 Conclusion

On real experimental extractant data with prospective validation, we show that a simple ensemble beats a published deep network, that the best active-learning acquisition depends on whether the goal is prediction or discovery (with no strategy dominating both), that active learning offers no predictive-accuracy advantage over random selection, and that uncertainty calibration is essential for trust but irrelevant to acquisition. These give practitioners an evidence-based decision guide and a reproducible benchmark for sample-efficient extractant discovery.

Data and Software Availability

The dataset is from ref. 1; all analysis code, splits, and figure- and number-generating scripts are released and archived (concept DOI 10.5281/zenodo.20764583), reproducing every value herein on a single laptop.

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Computations were run on a single Apple M4 laptop. The author conceived and directed the study, curated the data, made the scientific and methodological decisions, and verified

all results; in accordance with ACS policy, the author discloses that a generative AI assistant (Claude, Anthropic) was used under the author's direction for code implementation, experiment execution and analysis, and manuscript drafting. The author is accountable for the content.

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